Pokémon Types Classification

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Abstract

When a new Pokémon generation is unveiled, sprite previews are typically released to the public well before any combat statistics are disclosed. Fans often attempt to infer a Pokémon's type from these sprites alone, motivating the questions of whether automated models can perform the same inference reliably, and which Pokémon attributes carry the strongest type signal. This paper assembles a dataset of all 1025 Pokémon scraped from PokemonDB [4] and trains a series of classifiers over five distinct attribute sources: sprites, base stats, abilities, movesets, and Pokédex text entries. Classifiers are evaluated under three metrics—exact match, partial match, and a weighted match—on a held-out ten percent test set. Move-based models achieve the strongest performance, followed by the abilities model, while the stats and Pokédex-text models perform noticeably worse. An analysis of misclassifications suggests that sprite-based models tend to overpredict types because of spurious visual correlations, whereas Pokédex entries tend to emphasize only one facet of a dual-type Pokémon. Overall, the results indicate that a Pokémon's moveset is the most informative attribute for classifying its type, while sprites and Pokédex entries are less reliable due to their tendency to capture multiple aspects of a Pokémon simultaneously.

1 Introduction

When a new Pokémon generation is released, the most prominent Pokémon of that generation are typically showcased to the public early through video previews. These preview clips generally reveal only the sprite or model of the Pokémon, without disclosing any of its other attributes. To stay ahead of the competition, dedicated players often try to deduce additional information from the sprites alone, and the easiest piece of information to recover is the Pokémon's type. This project begins by using sprites to classify Pokémon types, and then expands to a broader study of other attributes in order to determine which one is the strongest predictor of type.

2 Related Work

Several prior projects have attempted to classify types from Pokémon sprites, but they often use more than one instance per Pokémon. For example, [1] uses every sprite from generations one through five, so a generation one Pokémon such as Mewtwo may contribute up to sixteen

different sprites drawn from the various games. Other implementations, such as [2], apply image augmentations to artificially increase the size of the dataset. Outside of sprite-based approaches, other work such as [3] uses stats together with non-deep-learning models to classify types.

3 Methodology

The data for this paper comes from the PokémonDB website [4]. The Pokédex page lists every Pokémon and its headline attributes, including Pokédex number, name, sprite, types, base stat total (BST), and stat breakdown. Each entry links to an individual Pokémon page, which contains further information such as height, weight, evolution line, evolution conditions, movesets, abilities, and flavor-text descriptions. All of the data from the individual pages is scraped and aggregated into a single JSON file. Ten percent of the data was randomly held out as a test set, and the remaining ninety percent was used for model training.

There are 18 types in total, and a Pokémon can have either one or two types depending on the species. Figure 1 shows the distribution of dual-typed Pokémon, with the right-most column reserved for Pokémon that have only a single type. Among the mono-typed Pokémon, water and normal types are the most common. Among the dual-typed Pokémon, the distribution is more spread out, with the largest being Normal + Flying types. Figure 2 treats each dual-typed Pokémon as two entries, and it shows that water, normal, and grass types have the highest overall occurrences, while ice types appear substantially less often than the second-lowest category.

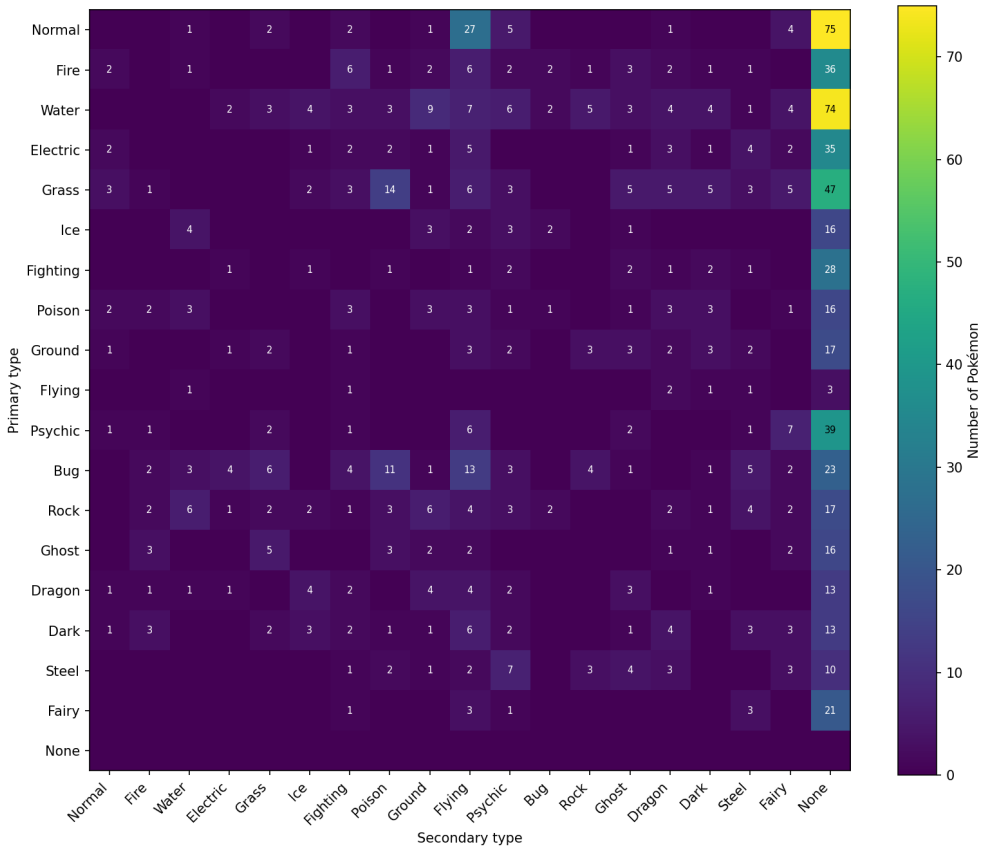

Figure 1: Distribution of type pairs across all Pokémon. Off-diagonal cells correspond to dual-typed Pokémon; the rightmost column corresponds to Pokémon with only a single type.

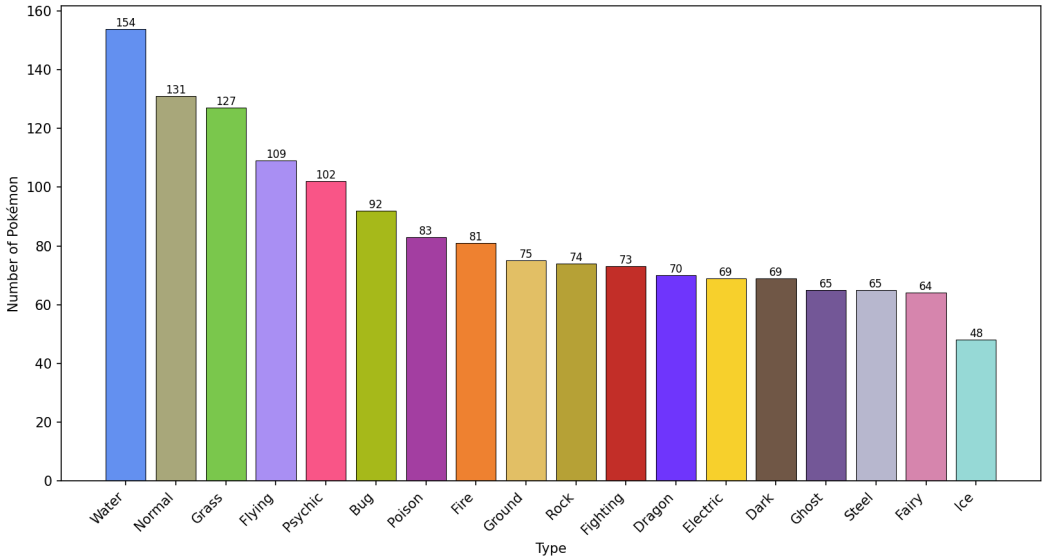

Figure 2: Total occurrences of each type, where dual-typed Pokémon contribute one count to each of their two types.

3.1 Sprites

Pokémon are often uniquely identified by their sprite. Although later games render Pokémon as 3D models, in-game screenshots can still be reduced to a 2D sprite-like image. The primary color of a sprite is strongly associated with the elemental types: red and orange for fire, blue for water and ice, yellow for electric, and green for grass. Texture is similarly correlated with material-based types, such as metallic surfaces for steel and blocky shapes for rock. Species-level features are also informative, including insect silhouettes for bug types, wings for flying types, and reptilian features for dragon types. Poses can also carry type information, such as combat stances for fighting types. Some examples are shown in Figure 3.

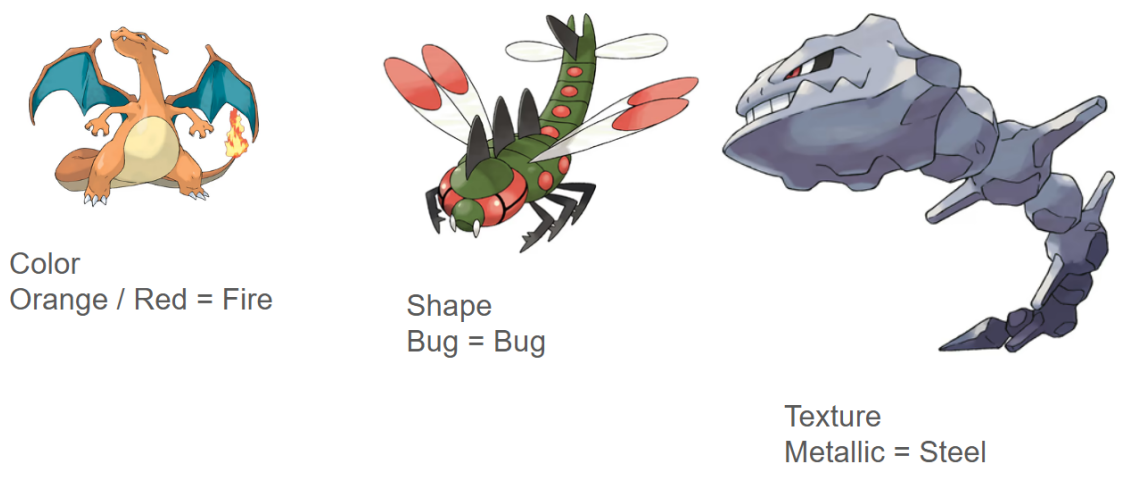

Figure 3: Examples of sprite-level features that are visually associated with particular types.

The sprite portion of the dataset consists of 1025 images, each of size 120×112. The first model is a convolutional neural network with two blocks, each consisting of convolution, normalization, ReLU, pooling, and dropout. Image augmentation is applied during training, including random horizontal flips, and random rotations up to 15 degrees.

This model produced poor results, likely because of its limited capacity and because it has to learn general image primitives such as edges and shapes entirely from scratch. To address this, a pretrained and more expressive model—ResNet18—was introduced. ResNet18 consists of four convolutional blocks, each containing two residual blocks with two convolutional layers apiece (for a total of sixteen convolutional layers), one input convolution, and a fully connected layer for the output. Residual connections are applied within each residual block, allowing the input to bypass two convolutions at a time.

3.2 Stats

Each Pokémon has a set of combat stats: health points, attack, defense, special attack, special defense, and speed. The sum of these stats is called the base stat total, or BST. Among the best-known correlations between stats and types are that bug types generally have lower stats, while dragon types have higher ones. A subcategory of Pokémon known as legendaries, who are the strongest Pokémon in their respective games, complicates stat-based

classification because most legendaries have a BST of 600 regardless of their type. This is partially offset, however, by the fact that most legendaries happen to be dragon types, while only one bug type is legendary. Figure 4 shows the average BST for each type, where dragon types have the highest mean and bug types the lowest.

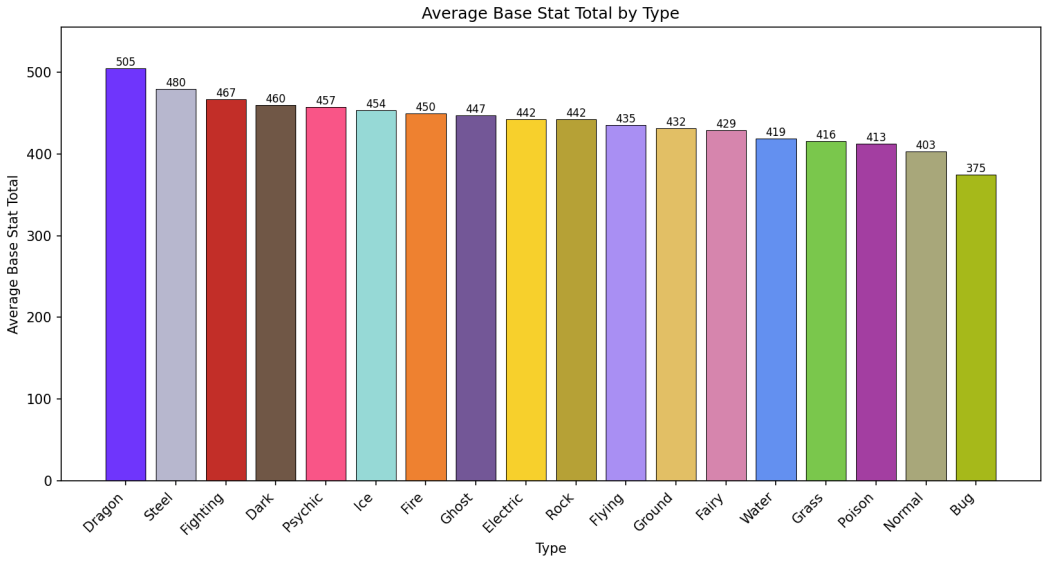

Figure 4: Average base stat total (BST) by type, where dual-typed Pokémon contribute their BST to each of their two types.

The model is a simple multilayer perceptron with a single input layer, one hidden layer, and an output layer. A small network is sufficient given that there are only six input features. However, the results of this model were poor, largely because of how limited the input features were. Pokémon have a concept of evolution, and higher evolution stages generally produce stronger Pokémon. This is problematic because a fully evolved bug type can have higher stats than a first-stage dragon type. The second attempt uses the same model architecture, but additionallyaugments the six base stats with additional numerical attributes such as height, weight, evolution stage, and several others, for a total of 29 features.

3.3 Abilities

There are currently nine Pokémon generations, each of which introduces a new batch of Pokémon and new mechanics. In generation three, Pokémon were assigned one or two abilities; in generation five, some Pokémon received an additional hidden ability. Abilities are assigned on a per-game basis rather than per-generation, so a generation one Pokémon reappearing in a generation five game is retroactively granted abilities (and possibly hidden abilities) in that game. As a result, every Pokémon now has at least one ability.

There are 274 unique abilities. A substantial portion of them belong to only one Pokémon—for example, Unseen Fist is exclusive to Urshifu, and Hadron Engine is exclusive to Miraidon. Other abilities were exclusive to a single Pokémon for a long time, until they were later

granted to newer Pokémon. For instance, Regigigas from generation four had Slow Start as its exclusive ability for four generations, until it was given to Vroom in generation nine.

The unique-assignment property of abilities makes it possible to roughly identify a Pokémon from its ability list alone, but sprites can accomplish the same thing, with arguably better results. What makes abilities particularly useful for type classification is the fact that certain abilities are restricted to Pokémon of a specific type. The most well-known examples are the starter abilities: Overgrow for grass, Torrent for water, and Blaze for fire. Examples of these abilities and their corresponding Pokémon are shown in Figure 5.

Figure 5: Examples of abilities and their associated Pokémon.

The models for the remaining attributes (abilities, moves, and Pokédex entries) follow a methodology similar to that used in large language models. Abilities are treated as tokens drawn from a vocabulary of size 274 (the number of unique abilities). The tokenized abilities are passed through an embedding layer to obtain an embedding vector, which is then processed by a multi-head self-attention block before being fed into a linear classifier.

3.4 Moves

Every Pokémon has a list of moves that it can learn for use in battle. Moves themselves carry a type, as well as accuracy and power values. In general, a Pokémon can only learn moves that match its own type, although there are notable exceptions: Pokémon with fists, for instance, can learn fist-based moves regardless of type. In addition, there are general-purpose moves such as Tackle that almost every Pokémon can learn.

When a Pokémon reaches a specific level, it learns moves from its level-up learnset. These moves are generally tailored to the Pokémon itself, and some are signature moves, which are unique to specific species. As with abilities, however, signature moves were later extended to other Pokémon in subsequent generations. In total, there are 622 level-up moves.

In addition to leveling up, Pokémon can be taught moves using Technical Machines, or TMs. Each TM teaches a single specific move, and the list of available TMs changes across generations. TM moves are more permissive than level-up moves and can generally be taught to many different Pokémon, although common-sense constraints still apply—for example, a Pokémon without fists cannot learn punching moves. There are 307 TM moves.

For these models, only the names of the moves are used as input. As with abilities, moves are tokenized, passed through an embedding layer, then through multi-head self-attention, and finally into a linear classifier. A separate model is trained for level-up moves and for TM moves.

3.5 Pokédex Entry

Each Pokémon has an extended textual description known as its Pokédex entry. Each game typically has its own unique Pokédex entry for a given Pokémon, but these entries usually differ from one another by only a few words. Some entries contain direct hints about the Pokémon's type, such as the following entry for the fire-type Charmander in Pokémon Yellow Version: "The flame at the tip of its tail makes a sound as it burns. You can only hear it in quiet places." Other entries describe the Pokémon's physiology without hinting at its type at all, such as the following entry for the dragon-type Goomy in Pokémon Y Version: "It's covered in a slimy membrane that makes any punches or kicks slide off it harmlessly."

For this model, the text of each entry from each game is extracted. When a Pokémon has multiple entries, those entries are concatenated into a single longer entry. This does not give a disproportionate advantage to Pokémon from earlier generations, because most entries are near-duplicates with only minor variations. The resulting text is encoded using Byte Pair Encoding with a vocabulary size of 2000. Once tokenized, the input is passed through an embedding layer and a multi-head self-attention layer before reaching a linear classifier.

4 Results

Each model outputs a vector whose elements represent the predicted probability that the Pokémon has each of the 18 types. A uniform threshold of 0.8 is applied across all models. Three metrics are reported. *Exact match* measures how often the model predicts exactly the correct set of types—one correct type for a mono-typed Pokémon, and both correct types for a dual-typed Pokémon. This is the strictest metric and is expected to yield the lowest values. *Partial match*, the least strict metric, measures how often the model predicts at least one of the correct types. *Weighted match* sits between the two: for a dual-typed Pokémon, the model receives full credit only if it predicts both types correctly, and half credit if it predicts only one of the two; for a mono-typed Pokémon, weighted match reduces to exact match. The total weighted match score is divided by the maximum possible number of points to yield a percentage.

Table 1 reports the results for each model.

Table 1: Classification performance of each model on the held-out test set.

Model	**Exact Match (%)**	**Partial Match (%)**	**Weighted Match (%)**
SpriteCNN	15.45	35.45	29.09
SpriteResNet18	28.18	53.64	40.91
Stats	1.82	2.73	2.27
UpdatedStats	11.82	22.73	17.27
Abilities	52.73	70.91	65.45
Level-up Moves	62.73	90.00	81.82
TM Moves	67.27	91.82	83.64
Pokédex	15.45	35.45	29.09

As expected, partial match is the highest-scoring metric for every model, since it is the most lenient. More informative, however, is the ratio between the stricter metrics and partial match, which is shown in Figure 6.

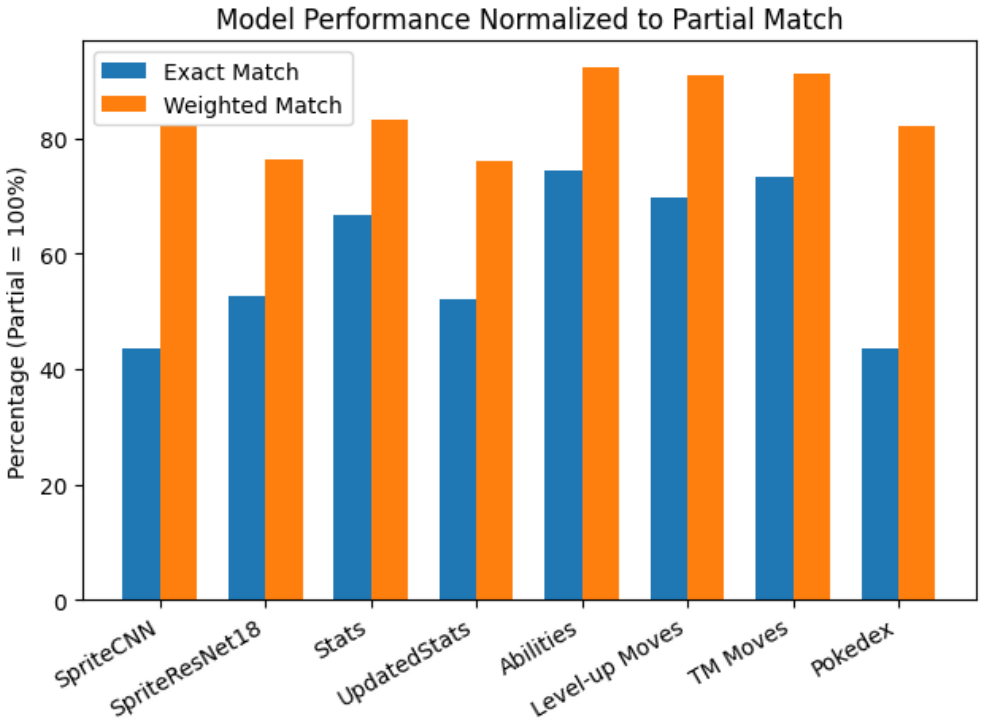

Figure 6: Ratio of exact match and weighted match scores to partial match scores across all models.

The Level-up Moves model yields the best results, but the Pokémon that it misclassifies are arguably more interesting. Table 2 lists several Pokémon that were misclassified by the Level-up Moves model but correctly classified by at least one other model. Table 3 lists several Pokémon that were misclassified by every model.

Table 2: Pokémon misclassified by the Level-up Moves model but correctly classified by another model.

Sprite	Pokémon	Actual Type	Predicted Type	Correct Model(s)
	Moltres	Fire, Flying	Fire, Flying, Normal	Abilities, TM Moves
	Poliwrath	Fighting, Water	Dark	TM Moves
	Umbreon	Dark	Normal	SpriteCNN

Table 3: Pokémon misclassified by all models.

Sprite	Pokémon	Actual Type	Predicted Type
	Dragalge	Dragon, Poison	Water
	Fezandipiti	Fairy, Poison	Flying, Normal
	Iron Boulder	Psychic, Rock	Steel

5 Discussion

Both stat-based models achieved the worst results, likely because stats are more closely tied to a Pokémon's anatomy than to its type. For example, slim, aerodynamic Pokémon tend to have high speed values, while large and bulky Pokémon tend to have high height, weight, and defense values, regardless of their type.

The next worst performers are the SpriteCNN and the Pokédex models, which produced comparable results, as shown in Table 1. Notably, these two models exhibit the lowest ratio of exact match to partial match in Figure 6, which is consistent with the nature of their inputs. Consider Moltres in Table 2: it is a fire bird, which makes it a Fire/Flying type, but most bird Pokémon are also classified as normal types because in generations one to seven, most Flying types were also Normal types. Sprites often contain more than one or two key aspects resuling in sprite-based models' tendency to overpredict types. The ResNet18

implementation mitigates this issue by having more layers to filter out the less relevant types but still suffers from the same underlying problem.

Pokédex entries exhibit the opposite failure, as they often focus on only one aspect of the Pokémon. For example, the Pokédex entry for Magcargo, a Rock/Fire type, emphasizes how hot it is but offers no hint that it is also a rock type.

The Abilities model sits in the middle of the rankings, which is reasonable given that there are a fair number of element-based abilities but also a non-negligible number of Pokémon-specific signature abilities.

The move-based models perform best, in part because each move itself has a type. Although the models are not explicitly trained on the moves' types, they appear to learn these types implicitly during training. The level-up learnset is slightly better than the TM learnset, most likely because level-up moves are more specifically tailored to each individual Pokémon.

More interesting than the aggregate model performance is an inspection of what the strongest models get wrong. Table 2 highlights several of the most illustrative misclassifications. Moltres and Umbreon are predicted to be Normal types because roughly two-thirds of their moves are either normal-type moves or generic moves. For example, Umbreon, aside from five normal-type moves, also learns Bite and Crunch, which are moves granted to any Pokémon with sharp teeth. Poliwrath is an outlier: it only learns three moves, but one of them is Bubble Beam, so at the very least the model should have predicted water as one of its types. In each of these cases, other models did manage to classify the Pokémon correctly.

There are also several Pokémon that none of the models were able to classify correctly, shown in Table 3. Dragalge resembles seaweed, and all of its observable attributes are water- or poison-based. Several models did manage to predict Poison, but none managed to predict Dragon. Dragalge is a Dragon type because it is based on the leafy seadragon, which is often depicted as a dragon in Japanese folklore; the models have no access to this background knowledge, and even most fans would not know this without being told. The other Pokémon in Table 3 follow a similar pattern: Fezandipiti is a bird that is not a Flying type, and Iron Boulder is a metallic beast that is not a Steel type.

To return to the original question, a Pokémon's moveset is the best attribute for classifying its type, followed by its abilities. Sprites capture too many different aspects of a Pokémon at once, which tends to lead to overclassification.

6 Acknowledgements

I would like to thank PokemonDB for providing publicly available data that made this research possible. I am also grateful to Professor Obradovic for offering a strong academic foundation that supported the completion of this project.

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